

RRC Procedure Timing under Handover Delay

MATLAB Code

```
clc;
clear;
close all;
% Number of observations
n = 10;
% RRC Parameters
handover_delay = randi([20 100],1,n); % ms
signaling_load = randi([10 50],1,n); % Messages
setup_time = randi([30 100],1,n); % ms
% Calculate throughput and failure rate
throughput = 100 - 0.5*handover_delay; % Mbps
failure_rate = handover_delay/10; % %
% Plot graphs
figure;
subplot(3,2,1);
plot(handover_delay, '-o');
title('Handover Delay');
ylabel('ms');
grid on;
subplot(3,2,2);
plot(signaling_load, '-o');
title('Signaling Load');
ylabel('Messages');
grid on;
subplot(3,2,3);
plot(setup_time, '-o');
title('Connection Setup Time');
ylabel('ms');
grid on;
subplot(3,2,4);
plot(throughput, '-o');
title('Throughput');
ylabel('Mbps');
grid on;
subplot(3,2,5);
plot(failure_rate, '-o');
title('Failure Rate');
ylabel('%');
grid on;
disp('Average Results');
disp(['Handover Delay = ', num2str(mean(handover_delay)), ' ms']);
disp(['Signaling Load = ', num2str(mean(signaling_load))]);
disp(['Setup Time = ', num2str(mean(setup_time)), ' ms']);
disp(['Throughput = ', num2str(mean(throughput)), ' Mbps']);
disp(['Failure Rate = ', num2str(mean(failure_rate)), ' %']);
```

Output

